

Dr. Laura Méndez Cuéllar

Evolutionary ecologist & macroecologist

Biodiversity of Plants · Department of Biology · Philipps-Universität Marburg
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Profile

Biologist specialising in evolutionary ecology and macroecology, with experience in plant micro-evolution, biogeography and phylogenetics. My research integrates spatial and genomic data to address questions about biodiversity patterns and evolutionary dynamics across time and space. I completed my PhD at iDiv Halle-Jena-Leipzig on the macroecological and micro-evolutionary consequences of seed dispersal limitation in Malagasy palms with large fruits, in relation to the recent extinction of megafrugivores on the island. I have built a molecular lab, planned and led fieldwork, supervised students and taught at master and graduate level. I work with large genomic and spatial datasets in R, Unix and on high-performance computing clusters.

Positions

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| 2025 – | Postdoctoral researcher and lecturer , Philipps-Universität Marburg
Biodiversity of Plants group (AG Zizka), Department of Biology. |
| 2024 – 2025 | Postdoctoral researcher , Helmholtz Centre for Environmental Research (UFZ), Halle
Flexpool project PaDRE – Patterns and Drivers of ecological marginality in Regional plant Extinctions.
Co-PIs: Sonja Knapp, Marten Winter, Erik Welk. |
| 2024 – 2025 | Parental leave |
| 2018 – 2024 | Doctoral researcher , iDiv Halle-Jena-Leipzig / University of Leipzig
PIs: Renske E. Onstein, Alexandra N. Müllner-Riehl. External reviewer: Anna Traveset. |
| 2016 – 2017 | Research assistant , National Museum of Natural Sciences (MNCN-CSIC), Madrid
Population genetics of freshwater and coastal fishes, the common midwife toad (<i>Alytes obstetricans</i>) and the brown spiny lobster (<i>Panulirus echinatus</i>). PIs: Annie Machordom, Jaime Bosch, Anabel Perdices. |
| 2015 | Research assistant , University of South-Eastern Norway, Bø i Telemark
Detection of <i>Anaplasma phagocytophilum</i> in <i>Ixodes ricinus</i> , and tick distribution along an altitudinal gradient. PI: Andrew Jenkins. |

Education

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| 2018 – 2023 | PhD in Biology (magna cum laude), University of Leipzig / iDiv
Signatures of the megafrugivore extinction on palms with large fruits in Madagascar. |
| 2016 – 2017 | MSc in Biodiversity and Conservation Biology , Pablo de Olavide University, Seville
Thesis at MNCN-CSIC Madrid: genetic structure and diversity of the Iberian populations of the freshwater blenny <i>Salaria fluviatilis</i> . |
| 2009 – 2015 | BSc in Biology , Autonomous University of Madrid
Erasmus year at the University of South-Eastern Norway (alpine biodiversity and environmental management). Thesis: how phytomining makes phytoremediation of heavy metals a profitable process. |

Publications

- 2025 Staude, I. R., Grenié, M., Thomas, C. D., Kühn, I., Zizka, A., Golivets, M., Ledger, S. E. H., Méndez, L. Many non-native plant species are threatened in parts of their native range. *New Phytologist*. doi.org/10.1111/nph.70193
- 2024 Méndez, L., Barratt, C. D., Durka, W., Kissling, W. D., Eiserhardt, W. L., Baker, W. J., Randrianasolo, V., Onstein, R. E. Genomic signatures of past megafrugivore-mediated dispersal in Malagasy palms. *Journal of Ecology* 112, 1583–1598. doi.org/10.1111/1365-2745.14340
- 2022 Zizka, A., Joerger-Hickfang, T., Imhof, S., Méndez, L. LiDAR sensors in smartphones can enrich herbarium specimens with 3D models of habitat at high precision and little cost. *Taxon* 72, 233–236. doi.org/10.1002/tax.12861
- 2022 Méndez, L., Viana, D. S., Alzate, A., Kissling, W. D., Eiserhardt, W. L., Rozzi, R., Rakotoarinivo, M., Onstein, R. E. Megafrugivores as fading shadows of the past. *Ecography* 2022(2). doi.org/10.1111/ecog.05885 – highlighted in *Nature Ecology & Evolution*.
- 2019 Méndez, L., Perdices, A., Machordom, A. Genetic structure and diversity of the Iberian populations of the freshwater blenny *Salaria fluviatilis* and its conservation implications. *Conservation Genetics* 20, 1409–1422. doi.org/10.1007/s10592-019-01205-0
- In prep. Méndez, L. et al. Insights into the demographic history of Malagasy palms · Méndez, L. et al. VascularPlantsNRL 1.0, the open vascular plants national Red Lists database · Pu, Y., Méndez, L. et al. Genetic legacy of megafauna seed dispersal in mainland Africa and Madagascar · Kambach, S. et al. Integrating national Red Lists and vegetation plot time series across Europe · Overcast, I. et al. Distribution of genetic diversity in ecological communities (IMEMEBA working group) · Garbowski, M. et al. Functional and phylogenetic approaches in ecological restoration (sFutures working group).

Teaching & supervision

- 2025 – Lecturer, Department of Biology, University of Marburg
- 2022 – 2024 Master and graduate courses, University of Leipzig and yDiv graduate school
Evolutionary drivers of macroecological patterns · Next generation sequencing (ddRADseq) · Estimating genetic diversity from short-read data, co-organised with C. D. Barratt.
- 2015 Phytoremediation, master module, University of South-Eastern Norway
- 2020 – 2023 Student supervision – Dimas E. Calderón Alfonso (master thesis, demographic models for Malagasy palms) and Adriane Vial (bachelor thesis, DNA barcoding of Malagasy palm samples), both as main supervisor with Renske E. Onstein.

Talks, outreach & service

- Talks Biogeography & GfÖ Macroecology (Marburg 2024) · GfÖ (Leipzig 2023) · GTÖ, invited symposium talk (České Budějovice 2023) · SFE² GfÖ EEF (Metz 2022) · EUNOPS (Geneva 2022, Paris 2019) · ECBC and Evolution (online 2021) · ATBC (Madagascar 2019) · invited seminars at the University of Amsterdam (2020) and the University of Leipzig (2019).
- Fieldwork Madagascar 2019, palm sampling with the Kew Madagascar Conservation Centre · Spain 2016, electrofishing · Norway 2015, small mammal trapping.

Outreach	Journal of Ecology blog (2024) · Long Night of Sciences, Leipzig (2021) · article in the University of Leipzig magazine (2020).
Awards	iDiv Science Communication Award (2022) · student travel award, ECBC Amsterdam (2020).
Service	Reviewer for <i>Biotropica</i> , <i>Global Ecology and Biogeography</i> and <i>Functional Ecology</i> .

Skills

Computing	Advanced R and Unix (bash), high-performance computing clusters, basic Python. Linear mixed-effects models, variation partitioning, species distribution models and ensembles, trait and genetic data imputation, GIS analyses. Structure, Bottleneck, BEAST, MrBayes, RAxML, Admixture, Plink, Stacks, BayesAss. Complex figures and maps in R.
Lab	DNA extraction, PCR and real-time PCR, gel electrophoresis, microsatellites, ddRADseq library preparation. Set-up and budgeting of a molecular lab.
Field	Planning and leading fieldwork, collecting herbarium specimens and DNA samples, identifying plant species in diverse environments.
Languages	Spanish (native), English (fluent), German, French (basic), Norwegian (basic).